



Parul University
Faculty of Engineering and Technology
Department of Applied Science & Humanities
Academic Year 2026-27
Subject: Quant and Reasoning (303105311)
Branch: CSE/ IT

TUTORIAL-5

21. The probability A getting a job is $\frac{1}{5}$ and that of B is $\frac{1}{7}$. What is the probability that only one of them gets a job?

- A. 1135 B. 1235
C. 27 D. 17

22. A letter is chosen at random from the word 'ASSASSINATION'. What is the probability that it is a vowel?

- A. 413 B. 813
C. 713 D. 613

23. A letter is chosen at random from the word 'ASSASSINATION'. What is the probability that it is a consonant?

- A. 413 B. 813
C. 713 D. 613

24. Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?

- A. 120 B. 920
C. 420 D. 14

25. One ball is picked up randomly from a bag containing 8 yellow, 7 blue and 6 black balls. What is the probability that it is neither yellow nor black?

- A. 13 B. 14
C. 12 D. 34

26. Two cards are drawn together from a pack of 52 cards. The probability that one is a club and one is a diamond?

- A. 1351 B. 152
C. 13102 D. 126

27. Two cards are drawn together at random from a pack of 52 cards. What is the probability of both the cards being Queens?

- A. 152 B. 1221
C. 2221 D. 126

28. Two dice are rolled together. What is the probability of getting two numbers whose product is even?

- A. 1736 B. 13
C. 34 D. 1125

29. When two dice are tossed, what is the probability that the total score is a prime number

- A. 14 B. 13
C. 23 D. 512

30. There are 10 prizes and 25 blanks in a lottery. If John has taken a lottery, what is the probability

for him to get a prize?

A. 47

B. 37

C. 27

D. 17

31. Six dice are tossed together. What is the probability of getting the same face in all the dice?

A. $\frac{1}{1765}$

B. $\frac{1}{165}$

C. $\frac{1}{166}$

D. $\frac{7}{64}$

32. Six dice are tossed together. What is the probability of getting different faces in all of the dice?

A. $\frac{1}{165}$

B. $\frac{1}{166}$

C. $\frac{6!}{165}$

D. $\frac{6!}{166}$

33. Six persons enter a lift on the ground floor of a nine floor apartment. Assuming that each of them independently and with equal probability can leave the lift at any floor beginning with the first, what is the probability that all the six persons are leaving the lift at different floors?

A. $\frac{8!}{8^6}$

B. $\frac{8!}{8^7}$

C. $\frac{8C6}{8^6}$

D. $\frac{8P6}{8^6}$

34. Raja has 4 tickets of a lottery for which 12 tickets were sold and 3 prizes are to be given. What is the probability that Raja will win at least one prize?

A. $\frac{4}{155}$

B. $\frac{4}{355}$

C. $\frac{12}{255}$

D. $\frac{14}{255}$

35. A basket contains 15 apples and 10 oranges out of which 4 apples and 2 oranges are defective. If a person takes two fruits at random, what is the probability that either both are apples or both are good

A. $\frac{22}{1300}$

B. None of these

C. $\frac{12}{12}$

D. $\frac{312}{401}$

36. A bag contains 4 blue, 5 white and 6 green balls. Two balls are drawn at random. What is the probability that both the balls are blue?

A. $\frac{2}{35}$

B. $\frac{1}{17}$

C. $\frac{1}{15}$

D. $\frac{2}{21}$

37. A bag contains 4 blue, 5 white and 6 green balls. Two balls are drawn at random. What is the probability that one ball is white?

A. $\frac{10}{21}$

B. $\frac{1}{2}$

C. $\frac{3}{4}$

D. $\frac{2}{35}$

38. Three boys P, Q and R are to speak at a function along with 5 others. If all of them speak in random order, what is the probability that P speaks before Q and Q speaks before R?

A. $\frac{2}{9}$

B. $\frac{1}{3}$

C. $\frac{1}{6}$

D. $\frac{1}{9}$

39. There is a point in a circle. What is the probability that this point is closer to its circumference than to the centre?

A. $\frac{1}{3}$

B. $\frac{1}{4}$

C. $\frac{1}{2}$

D. $\frac{3}{4}$

40. If 10 letters are to be placed in 10 addressed envelopes, then what is the probability that at least

one letter is placed in wrong addressed envelope?

a.1/10!

b.1/9!

c.1- (1/10!)

d.9/10